Package 'prime'

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Type Package

Title Proportional Inclusive Marginal Effects

Version 0.5.2.9101

Description Estimates proportional global variable importance for noisy input-response data with dependent predictors. A response-relevant screen adapts 'FIRST' from Huang and Joseph (2025) <doi:10.1080/00401706.2025.2483531> using local-linear residualization, a Bayesian-information-criterion-gated ridge start, and query-calibrated forward and backward comparisons. Proportional allocation follows Herin et al. (2024) <doi:10.1137/22M153032X>. Exact nearest-neighbor searches use the full sample by default; deterministic 'Data Twinning' from Vakayil and Joseph (2022) <doi:10.1002/sam.11574> can reduce the outer query average.

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Encoding UTF-8

URL https://github.com/ChoYoungSeo/prime-importance-r

BugReports https://github.com/ChoYoungSeo/prime-importance-r/issues

Depends R (>= 4.2.0)

Imports FNN, MASS, parallel, Rcpp

LinkingTo Rcpp, RcppArmadillo

Suggests testthat (>= 3.0.0), twinning

Config/testthat/edition 3

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`prime-package` *PRIME: Proportional Inclusive Marginal Effects*

Description

The **prime** package implements the proportional inclusive marginal effects of Cho and Joseph (2026), which estimates model-free variable importance from noisy input-output data. It also provides model-free variable selection.

Details

PRIME estimates explained-variance subset values with local-linear neighborhoods, constructs a BIC-gated ridge-start candidate set, applies paired querywise forward and backward calibration, and evaluates proportional marginal effects from noise-adjusted risks on the retained coordinates.

The main interface is `prime()`, which runs coordinate selection, subset-value estimation, and proportional allocation. The supporting interface is `prime_screen()` for coordinate selection and diagnostic traces without the allocation step. Use `summary(fit)` or `summary(screen)` for a concise result summary, and `plot(fit)` to display variable importance.

Exact subset enumeration grows exponentially with the retained dimension. The ambient dimension affects screening, while the selected dimension determines the final allocation cost. Data Twinning (Vakayil and Joseph, 2022) can reduce the query-average cost for large datasets. Fitted objects have concise print, summary, and plot methods.

Acknowledgments

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Author(s)

Youngseo Cho and V. Roshan Joseph

References

Herin, M., Il Idrissi, M., Chabridon, V., and Iooss, B. (2024). Proportional marginal effects for global sensitivity analysis. *SIAM/ASA Journal on Uncertainty Quantification*, 12(2), 667–692. doi:10.1137/22M153032X.

Cho, Y. and Joseph, V. R. (2026). Measuring Variable Importance Using Proportional Inclusive Marginal Effects. arXiv:2610.17053. doi:10.48550/arXiv.2610.17053.

Vakayil, A. and Joseph, V. R. (2022). Data Twinning. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, 15, 598–610. doi:10.1002/sam.11574.

See Also

`prime`, `plot.prime_fit`, `prime_screen`

`plot.prime_fit` *Plot PRIME Variable Importance*

Description

Displays PRIME importance from a fitted result. Retained coordinates are highlighted and excluded coordinates remain visible at zero.

Usage

```
## S3 method for class 'prime_fit'
plot(x, y = NULL, max_display = 20L,
  main = NULL, xlab = "PRIME importance",
  col = "#087F8C", zero_col = "#A5ADB4", ...)
```

Arguments

`x`	A fitted `prime_fit` object returned by `prime()`.
`y`	Unused.
`max_display`	Maximum number of coordinates to display.
`main`	Optional plot title.
`xlab`	Horizontal-axis label.
`col`	Color for retained coordinates.
`zero_col`	Color for excluded coordinates.
`...`	Additional graphical parameters passed to `plot.default()`.

Value

The fitted object, invisibly.

See Also

`prime`

Examples

```
set.seed(5)
X <- matrix(runif(960, -1, 1), nrow = 120)
colnames(X) <- paste0("X", seq_len(ncol(X)))
y <- X[, 1] - X[, 2] + X[, 5]^2 + rnorm(120, sd = 0.08)

fit <- prime(X, y)
plot(fit)
```

`prime` *Proportional Inclusive Marginal Effects*

Description

Estimates Proportional Inclusive Marginal Effects of Cho and Joseph (2026).

Usage

```
prime(X, y, rescale = TRUE, twin_r = NULL, lambda = 1e-6)
```

Arguments

`X`	Numeric predictor matrix or data frame with unique column names.
`y`	Finite numeric response vector with length `nrow(X)`.
`rescale`	Logical; standardize predictors before distance calculations.
`twin_r`	Optional integer inverse Data-Twinning split ratio. `NULL` uses every observation as a query location.
`lambda`	Nonnegative slope-only numerical nugget used in local-linear fits.

Details

The function uses BIC-gated ridge-start screening, query-calibrated forward and backward reduction, and proportional allocation (Herin et al., 2024) on the retained total-effect set function. See Cho and Joseph (2026) for details.

For retained dimension d, the local neighborhood size is

$$K_d = \left\lceil \max\{1.5\sqrt{n},\, 3d\} \right\rceil,$$

subject to the local-fit feasibility cap. The local nugget is applied to slopes only. Every observation is used as a query by default; Data Twinning reduces the outer query average without removing observations from the neighbor pool.

Exact allocation evaluates 2^d subset values and therefore scales exponentially with the retained dimension.

Use `summary(fit)` to see retained coordinates, importance estimates, explained variation, and projection diagnostics.

Value

An object of class `prime_fit` with components:

- `call`: matched function call;
- `effects`: full-length, zero-padded PRIME allocation;
- `reduced_effects`: allocation on retained coordinates;
- `selected` and `selected_names`: retained coordinates;
- `screen`: complete `prime_screen` result;
- `subset_values`: raw and projected subset values with complement and noise-adjusted risks;
- `signal`, `noise_floor`, and `risk_full`: estimated explained variation and full retained-set risk;
- `projection`: monotone-projection diagnostics;
- `settings`: fixed numerical and statistical settings.

References

Herin, M., Il Idrissi, M., Chabridon, V., and Iooss, B. (2024). Proportional marginal effects for global sensitivity analysis. *SIAM/ASA Journal on Uncertainty Quantification*, 12(2), 667–692. doi:10.1137/22M153032X.

Cho, Y. and Joseph, V. R. (2026). Measuring Variable Importance Using Proportional Inclusive Marginal Effects. arXiv:2610.17053. doi:10.48550/arXiv.2610.17053.

See Also

plot.prime_fit, prime_screen

Examples

```
set.seed(5)
X <- matrix(runif(960, -1, 1), nrow = 120)
colnames(X) <- paste0("X", seq_len(ncol(X)))
y <- X[, 1] - X[, 2] + X[, 5]^2 + rnorm(120, sd = 0.08)

fit <- prime(X, y)
fit$selected_names
round(fit$effects, 3)
summary(fit)
plot(fit)
```

`prime_screen` *Nonparametric Variable Selection*

Description

Implements the nonparametric model-free variable selection technique in Cho and Joseph (2026).

Usage

```
prime_screen(X, y, rescale = TRUE, twin_r = NULL, lambda = 1e-6)
```

Arguments

`X`	Numeric predictor matrix or data frame with unique column names.
`y`	Finite numeric response vector with length `nrow(X)`.
`rescale`	Logical; standardize predictors before distance calculations.
`twin_r`	Optional integer inverse Data-Twinning split ratio. `NULL` uses every observation as a query location.
`lambda`	Nonnegative slope-only numerical nugget used in local-linear fits.

Details

The ridge ordering is searched through $\lceil 1.5 \log\{\max(2, p)\}\rceil$ coordinates, including the intercept-only model, and BIC selects the initial prefix. The local path adds the coordinate with the largest explained-variance gain when its paired querywise calibrated gain is positive. Backward reduction sequentially removes zero calibrated deletion losses and recomputes all projected neighborhoods after every removal.

Use `summary(screen)` to see the ridge seed, forward path, retained coordinates, and backward-reduction counts.

Value

An object of class `prime_screen` with components:

- `call`: matched function call;
- `candidates` and `candidate_names`: inclusive forward set;
- `selected` and `selected_names`: backward-cleaned set;
- `forward`: ridge seed, local path, stopping reason, and trace;
- `backward`: retained set and deletion-comparison trace;
- `settings`: neighborhood, query, nugget, and calibration rules.

References

Cho, Y. and Joseph, V. R. (2026). Measuring Variable Importance Using Proportional Inclusive Marginal Effects. arXiv:2610.17053. doi:10.48550/arXiv.2610.17053.

See Also

`prime`

Examples

```
set.seed(5)
X <- matrix(runif(960, -1, 1), nrow = 120)
colnames(X) <- paste0("X", seq_len(ncol(X)))
y <- X[, 1] - X[, 2] + X[, 5]^2 + rnorm(120, sd = 0.08)

screen <- prime_screen(X, y)
screen$selected_names
summary(screen)
screen$forward$trace
screen$backward$trace
```

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